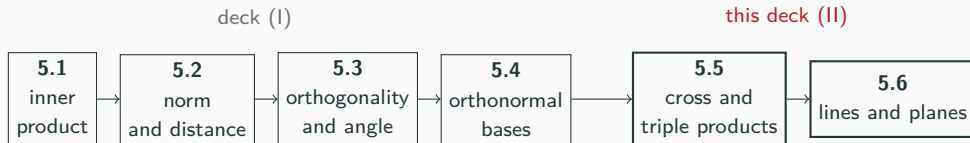


Linear Algebra

5 Inner product and geometry (II)

José Firmino Aguilar Madeira

Chapter map



Central idea: in $\mathbb{R}^3$, the inner product (deck I) and determinants (Ch. 4) give a complete geometry: areas, volumes, lines and planes. Relative positions are decided by the rank (Ch. 3).

Exercises: Sheet 5, Ex. S1–S10 (new exercises, after Ex. 98).

5.5 Cross product and scalar triple product

Cross product

Definition (only in $\mathbb{R}^3$)

For $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$, the **cross product** is the vector

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = (u_2 v_3 - u_3 v_2, \quad u_3 v_1 - u_1 v_3, \quad u_1 v_2 - u_2 v_1).$$

The 'determinant' is formal (its 1st row contains vectors): use the Laplace expansion along the 1st row (Ch. 4).

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The 'determinant' is formal (its 1st row contains vectors): use the Laplace expansion along the 1st row (Ch. 4). **Example.** $(1, 2, 0) \times (0, 1, 3) = (2 \cdot 3 - 0 \cdot 1, \quad 0 \cdot 0 - 1 \cdot 3, \quad 1 \cdot 1 - 2 \cdot 0) = (6, -3, 1).$

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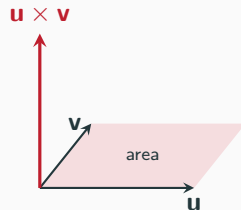
Check: $(6, -3, 1) \cdot (1, 2, 0) = 0$ and $(6, -3, 1) \cdot (0, 1, 3) = 0$.

Properties of the cross product

1. $\mathbf{u} \times \mathbf{v}$ is orthogonal to $\mathbf{u}$ and to $\mathbf{v}$;
2. $\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$ (swapping two rows, Ch. 4);
3. $\mathbf{u} \times \mathbf{v} = \mathbf{0} \iff \mathbf{u}, \mathbf{v}$ are linearly dependent;
4. $\mathbf{u} \times (\alpha \mathbf{v} + \beta \mathbf{w}) = \alpha \mathbf{u} \times \mathbf{v} + \beta \mathbf{u} \times \mathbf{w}$;
5. $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta = \text{area of the parallelogram}.$

The orientation of $\mathbf{u} \times \mathbf{v}$ is given by the right-hand rule:

$$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3, \quad \mathbf{e}_2 \times \mathbf{e}_3 = \mathbf{e}_1, \quad \mathbf{e}_3 \times \mathbf{e}_1 = \mathbf{e}_2.$$



Properties of the cross product

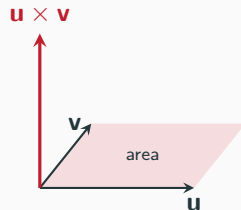
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Caution

It is neither commutative nor associative: $(\mathbf{e}_1 \times \mathbf{e}_1) \times \mathbf{e}_2 = \mathbf{0}$, but $\mathbf{e}_1 \times (\mathbf{e}_1 \times \mathbf{e}_2) = \mathbf{e}_1 \times \mathbf{e}_3 = -\mathbf{e}_2$.



Parallelogram defined by $\mathbf{u}$ and $\mathbf{v}$: $\text{area} = \|\mathbf{u} \times \mathbf{v}\|$.

Triangle ABC : $\text{area} = \frac{1}{2} \|\overrightarrow{AB} \times \overrightarrow{AC}\|$.

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Link to Ch. 4. The parallelogram spanned by the columns $(3, 1)$ and $(1, 2)$, placed in the plane $z = 0$:

$$(3, 1, 0) \times (1, 2, 0) = (0, 0, 3 \cdot 2 - 1 \cdot 1) = (0, 0, 5), \quad \text{area} = 5 = \left| \det \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \right|.$$

In the plane, the area via the cross product and via the determinant is the same computation.

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In the plane, the area via the cross product and via the determinant is the same computation.

Useful identity: $\|\mathbf{u} \times \mathbf{v}\|^2 = \|\mathbf{u}\|^2 \|\mathbf{v}\|^2 - (\mathbf{u} \cdot \mathbf{v})^2$ (it links the two products; Ex. S3).

Scalar triple product and volume

Definition

The **scalar triple product** of $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ is the number

$$[\mathbf{u}, \mathbf{v}, \mathbf{w}] = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}.$$

- $|[\mathbf{u}, \mathbf{v}, \mathbf{w}]| = \text{volume of the parallelepiped}$ defined by $\mathbf{u}, \mathbf{v}, \mathbf{w}$.
- $[\mathbf{u}, \mathbf{v}, \mathbf{w}] = 0 \iff \mathbf{u}, \mathbf{v}, \mathbf{w}$ are **coplanar** (linearly dependent, Ch. 2).

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Example (this course's running matrix). With the columns of C , $\mathbf{c}_1 = (2, -6, -6)$, $\mathbf{c}_2 = (0, 2, 0)$, $\mathbf{c}_3 = (0, -3, -1)$:

$$\mathbf{c}_2 \times \mathbf{c}_3 = (-2, 0, 0), \quad [\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3] = \mathbf{c}_1 \cdot (-2, 0, 0) = -4 = \det C.$$

Volume 4: this is the 'C multiplies volumes by 4' of Ch. 4.

- **Ex. S1** cross product, orthogonality and area. Compare $\mathbf{u} \times \mathbf{v}$ with $\mathbf{v} \times \mathbf{u}$.
- **Ex. S2** area of a triangle given by three points.
- **Ex. S3** properties: write the computations in coordinates.
- **Ex. S4** volume and the condition for coplanarity.

5.6 Lines and planes in $\mathbb{R}^3$

A line is determined by a point $P_0 = (x_0, y_0, z_0)$ and a **direction vector** $\mathbf{d} = (d_1, d_2, d_3) \neq \mathbf{0}$.

- **Vector equation:** $(x, y, z) = P_0 + t\mathbf{d}, \quad t \in \mathbb{R}.$
- **Parametric equations:** $x = x_0 + td_1, \quad y = y_0 + td_2, \quad z = z_0 + td_3.$
- **Cartesian equations** (if $d_1, d_2, d_3 \neq 0$): $\frac{x - x_0}{d_1} = \frac{y - y_0}{d_2} = \frac{z - z_0}{d_3}.$

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Example. Line through $A(1, 0, 2)$ and $B(3, 1, 0)$: $\mathbf{d} = \overrightarrow{AB} = (2, 1, -2),$

$$(x, y, z) = (1, 0, 2) + t(2, 1, -2), \quad \frac{x - 1}{2} = y = \frac{z - 2}{-2}.$$

Compare with Ch. 3: the solutions of a system with infinitely many solutions and $g = 1$ are $\mathbf{x}_p + t\mathbf{v}$ — a line.

A plane is determined by a point P_0 and a **normal vector** $\mathbf{n} = (a, b, c) \neq \mathbf{0}$:

$$X \in \pi \iff \mathbf{n} \cdot (X - P_0) = 0 \iff \boxed{ax + by + cz = d}, \quad d = \mathbf{n} \cdot P_0.$$

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Plane through three points A, B, C (not collinear): $\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC}$.

Example. $A(1, 0, 0), B(0, 1, 0), C(0, 0, 1)$: $\overrightarrow{AB} = (-1, 1, 0), \overrightarrow{AC} = (-1, 0, 1),$

$$\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC} = (1, 1, 1), \quad \pi : x + y + z = 1.$$

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There is also the vector equation $X = P_0 + s\mathbf{u} + t\mathbf{v}$ ($\mathbf{u}, \mathbf{v}$ lin. indep.), and $\mathbf{n} = \mathbf{u} \times \mathbf{v}$ converts one form into the other.

Relative position = classifying a system (Ch. 3)

Two planes π_1, π_2 : the system of the two equations, with $r = \text{car } A$ and $s = \text{car}[A|\mathbf{b}]$:

- $r = s = 2$: they intersect in a **line** (infinitely many solutions, $g = 1$), with direction $\mathbf{n}_1 \times \mathbf{n}_2$;
- $r = 1 < s = 2$: distinct **parallel** planes (inconsistent); $r = s = 1$: **coincident**.

Three planes with $\text{car } A = 3$: a single common point (unique solution).

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Example. $x + y + z = 1$ and $x - y = 0$: $x = y$, $z = 1 - 2x$, so the line $(0, 0, 1) + t(1, 1, -2)$. Check: $(1, 1, 1) \times (1, -1, 0) = (1, 1, -2)$.

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Line and plane: substituting the parametric equations into the plane's equation gives an equation in t : one solution (**intersecting**), none (**parallel**), every t (**contained**). In the last two cases, $\mathbf{d} \cdot \mathbf{n} = 0$.

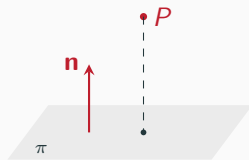
Distance from a point to a plane

Formula

For $P = (x_0, y_0, z_0)$ and $\pi : ax + by + cz = d$:

$$d(P, \pi) = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Where it comes from: it is the length of the projection of $\overrightarrow{AP}$ onto $\mathbf{n}$ ($A \in \pi$), as in 5.4.



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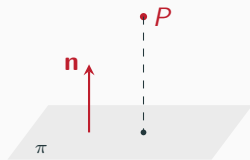
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Example. $P(2, 2, 2)$, $\pi : x + y + z = 1$: $d(P, \pi) = \frac{|2 + 2 + 2 - 1|}{\sqrt{3}} = \frac{5}{\sqrt{3}}.$



- **Ex. S5** equations of a line; checking whether a point lies on it.
- **Ex. S6, S7** equation of a plane: from the normal vector; through three points.
- **Ex. S8** relative position of planes via the rank (Ch. 3).
- **Ex. S9** line and plane: the three possible cases.
- **Ex. S10** distance from a point to a plane and the closest point of the plane.

3D graphics. Surfaces are made of millions of triangles ABC . For each one:

$$\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC} \quad (\text{normal}), \quad \text{brightness} \propto \cos \theta = \frac{\mathbf{n} \cdot \ell}{\|\mathbf{n}\| \|\ell\|} \quad (\ell: \text{light direction}).$$

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Mechanics. The **moment** (torque) of a force $\mathbf{F}$ applied at the point with position vector $\mathbf{r}$ is $\mathbf{M} = \mathbf{r} \times \mathbf{F}$: its direction is the axis of rotation and its norm, $\|\mathbf{r}\| \|\mathbf{F}\| \sin \theta$, measures the turning effect (which is why a longer spanner makes a bolt easier to tighten).

Quick reference — Chapter 5 (II)

Concept	How to obtain it / what it says
Cross product	$\mathbf{u} \times \mathbf{v}$: formal determinant with $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ in the 1st row
Properties	$\perp \mathbf{u}, \mathbf{v}$; $\mathbf{v} \times \mathbf{u} = -\mathbf{u} \times \mathbf{v}$; $= \mathbf{0} \iff$ lin. dep.; not associative
Area	parallelogram $\ \mathbf{u} \times \mathbf{v}\ $; triangle $\frac{1}{2}\ \overrightarrow{AB} \times \overrightarrow{AC}\ $
Triple product	$[\mathbf{u}, \mathbf{v}, \mathbf{w}] = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \det$; volume $= [\mathbf{u}, \mathbf{v}, \mathbf{w}] $
Coplanar	$[\mathbf{u}, \mathbf{v}, \mathbf{w}] = 0$
Line	$P_0 + t\mathbf{d}$; parametric; Cartesian
Plane	$\mathbf{n} \cdot (X - P_0) = 0$, $ax + by + cz = d$; through 3 points: $\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC}$
Relative position	classify the system by its rank (Ch. 3)
Line and plane	equation in t : one, no or infinitely many solutions
Distance to a plane	$ ax_0 + by_0 + cz_0 - d /\sqrt{a^2 + b^2 + c^2}$

KEY TAKEAWAYS — Chapter 5 (II)

1. The cross product gives a vector orthogonal to both given vectors, with norm equal to the area of the parallelogram.
2. The scalar triple product is a determinant: its absolute value is a volume, and it is zero when the vectors are coplanar.
3. A line is a point plus the multiples of a direction vector; a plane is given by a point and a normal vector.
4. Relative positions of lines and planes are decided by classifying a system (Ch. 3).
5. The distance from a point to a plane is a projection onto the normal vector.

Next (Ch. 6): eigenvalues and eigenvectors — the directions that a matrix only stretches or shrinks.

References (full list: course bibliography)

- Santana, A. P., Queiró, J. F., *Introdução à Álgebra Linear*, Gradiva (in Portuguese).
- Lipschutz, S., Lipson, M., *Schaum's Outline of Linear Algebra*, McGraw-Hill — many worked exercises.
- Strang, G., *Introduction to Linear Algebra*, Wellesley-Cambridge Press.

Resources — Sheet 5 (exercises 83–98 and S1–S10) and its solutions.